3 A technical overview of contract testing

This chapter covers

- Defining consumers, providers, contracts, and contract brokers
- Contract testing life cycle
- Contract testing tools
- Contract testing approaches
- Communication types supported by contract testing

Before we walk through the implementation steps of contract tests (in later chapters), we'll first introduce the terminology and tooling. This chapter discusses what consumers and providers are in relation to software development. In addition, before writing contract test code, you must start conversations with key stakeholders regarding where to store the contracts.

In this chapter, we explain the core contract testing concepts. Then, we talk about deciding which contract testing approach to take. We cover the two approaches offered within the contract testing domain: consumer-driven contract testing (CDCT), often referred to as the traditional approach, and provider-driven contract testing (PDCT). We describe the pros and cons of CDCT and PDCT, guiding you to the best approach in your context.

Throughout the chapter, we compare contract testing to integration testing; therefore, a basic understanding or previous experience in the integration testing area is beneficial. Even though contract testing doesn't replace integration testing completely, comparing the two helps us understand the fundamental concepts of contract testing. Lewis Prescott (one of the authors) describes contract testing as flipping integration testing upside down, like reversing the integration testing process. The usual approach within integration testing is for the service team, for example, the API service, to implement the tests during the development process. Once the